

AI Study Planner for Engineering Students

Build the Future of Planning Your Study Schedule

THE CHALLENGE:

Engineering students face a uniquely demanding academic environment. They juggle multiple technically intensive subjects at once—each with different prerequisites, assessment styles, and cognitive demands.

Traditional study planning methods like static calendars, to-do lists, or generic productivity apps fail to adapt to the dynamic, interconnected, and high-cognitive-load nature of engineering coursework.

As a result, students often study hard—but not smart.

PROBLEM STATEMENT:

Engineering students struggle to efficiently allocate limited study time across complex subjects while maintaining consistent progress, meeting deadlines, and achieving deep conceptual understanding.

This challenge is driven by:

- Cognitive Load Imbalance – Different subjects demand different levels of mental effort, but students lack systems to plan high-focus vs low-focus study sessions effectively.
- Prerequisite Dependencies – Courses build heavily on prior concepts, yet students often fail to identify foundational gaps blocking progress.
- Dynamic Prioritization Issues – Deadlines, exams, and unexpected topic difficulties constantly shift priorities, making manual planning ineffective.
- Inefficient Study Patterns – Cramming, shallow practice, and last-minute studying lead to poor retention and high stress.
- Lack of Personalization – Existing tools don't adapt to individual learning speed, weak subjects, preferred study times, or personal constraints.

YOUR MISSION:

Design and develop an AI-powered study planner tailored specifically for engineering students that:

- Analyzes subjects, deadlines, prerequisites, and cognitive load
- Creates personalized, adaptive study schedules
- Helps students balance deep learning with timely completion
- Evolves dynamically as priorities, performance, and difficulty change

INPUT FROM USER:

1. Student Details like Name, College, Graduation Year, Email ID
2. Subjects and their respective credits for which study plan needs to be made.
3. Number of study hours available per day during weekdays and weekends (can be collected separately from user)
4. Target date until which the subjects need to be finished.
5. Any strong and weak areas of each subject, if any, along with Self-Rated Confidence Level (1 – 5) for each subject.

Sample Input:

Student Details

Name: Aman

College: XYZ Institute of Technology

Branch: Computer Science Engineering

Graduation Year: 2026

Email ID: aman@example.com

Subjects & Credits

1. Data Structures – 4 Credits
2. Operating Systems – 3 Credits
3. Engineering Mathematics – 4 Credits

Study Time Availability

Weekdays: 3 hours/day

Weekends: 6 hours/day

Preferred Study Time: Night

Target Completion Date

Target Completion Date: 15 March 2026

Data Structures

Strong Areas: Arrays, Linked Lists

Weak Areas: Trees, Graphs

Confidence Level: 3/5

Operating Systems

Strong Areas: Processes, Threads

Weak Areas: Deadlocks, Memory Management

Confidence Level: 2/5

Engineering Mathematics

Strong Areas: Differential Equations

Weak Areas: Laplace Transform

Confidence Level: 3/5

Sample Output – Expected Direction (Guidance)

The generated output should feel **actionable, intelligent, and motivating**, not just a static timetable.

A strong solution may include:

1. Visual, Easy-to-Scan Schedule

- Daily or weekly calendar view
- Color-coded subjects based on **cognitive load** (high / medium / low)
- Clear distinction between **learning, practice, revision, and buffer time**

2. Subject-Wise Focus Breakdown

- Percentage or hour-based allocation per subject
- Justification such as:
 - *“More time allocated due to low confidence and higher credits”*
 - *“Reduced load for strong topics to avoid over-studying”*

3. Smart Prioritization Logic

- Weak and prerequisite-heavy topics scheduled **earlier**
- High-focus topics placed in **preferred study hours**
- Lighter tasks (revision, notes) placed in low-energy slots

4. Actionable Next Steps

Instead of generic advice, outputs should include:

- “Next 7 days focus: Deadlocks, Graph Traversals”
- “Revise Trees before starting Graphs to close prerequisite gap”
- “Complete Laplace Transform before Week 3 to avoid backlog”

5. Progress & Adaptation Indicators

- Weekly confidence checkpoints
- Automatic rebalancing suggestions such as:
 - “Confidence in OS improved from 2 → 3; reallocate 30 mins to DS”
- Buffer time clearly marked for spillovers

6. Outcome-Oriented Summary

End the output with clarity:

- Estimated completion timeline
 - Expected confidence improvement
 - Reduction in last-minute workload
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DELIVERABLES:

- Working Prototype – A functional MVP that generates personalized study plans
- Adaptive Scheduling Logic – Accounts for subject difficulty, deadlines, and prerequisite gaps
- Actionable Insights – Clear guidance on *what to study, when, and why*
- User-Friendly Interface – Clean, intuitive design for daily student use
- Demo/Presentation – 5-minute video explaining the problem, solution, and impact

EVALUATION CRITERIA:

- **Impact (20%)** –How effectively the solution assesses student inputs and generates a personalized, realistic study calendar within 2 minutes of user effort.
- **Innovation (20%)** – Is the approach creative and unique?
- **Technical Execution (20%)** – Code quality and functionality including ReadMe file quality.
- **User Experience (25%)** – Ease of input, clarity of output schedules, and overall usability for daily student use (**hosted solutions will receive bonus consideration**).
- **Presentation (15%)** – Quality of demo video and explanation.

WHO WINS?

The Individual that builds the most practical, personalized, and impactful AI study planner—one that genuinely helps engineering students study smarter, not harder as per the evaluation criteria mentioned above.

Ready to re-engineer the way students learn? Let's build it.

Submission Guidelines:

- Users must submit the **link to their GitHub repository** on the competition portal (**Unstop**, where you completed your hackathon registration).
- The **GitHub repository must be public**. Private repositories will lead to **disqualification**.
- Hosting the project is optional but will give you an edge during evaluation.
- **Mandatory:** Include a **screen-recording video of your live, working project** in the **README.md** of your GitHub repository.

FAQs

Q. What Tech Stack to use?

A. You can use any tech stack as per your expertise.

Q. Can I use AI to build the solution?

A. Yes you can use AI Tools like ChatGPT, Emergent, Co-Pilot etc. to build the solution for rapid prototyping. But make sure you understand the platform that you have built.

Q. Will I get a certificate?

A. Yes, but only if you have done a Valid Submission.

Q. When will the results be announced?

A. The results will be announced within a week's time of the hackathon submission.

Q. When will the certificates be given?

A. Certificates will be issued only to Valid submissions defined as public gitHub Repository and a working prototype as per the problem statement. The certificates will be issued within 7 Days of the hackathon.

Q. How will I get the certificate?

A. You will get a digital verifiable certificate via your email within 7 days of the hackathon.

Q. How much time do I have to make the submission?

A. Submission Deadline is **7th Feb, 2026 6:30 PM**. You can check Unstop Portal.



Q. I have more doubts, how can I contact the UnsaidTalks Team?

A. You can reach out @ +91-7303573374 on whatsapp or @ info@unsaidtalks.com if you have any doubts or feedback for us.

Q. Where do I have to submit the solution?

A. You need to upload the code in your public github repository and submit the link on the unstop portal where you did registration.

Q. Who is organizing this Hackathon?

A. This hackathon is organised by UnsaidTalks Education (www.unsaidtalks.com) - India's Leading Mentorship Platform helping 1000s of students get Interview ready and helping them crack internships and jobs via network of Industry Professionals from Google, Amazon, Microsoft and other Top Tech Companies. Currently admissions are open for April '2026 batch for Build4Hire Program. Details can be found here: <https://unsaidtalks.com/build4hire-webdev-genai/>